MECHATRONICS AND MEMS

Subject Code : 23A03702d

UNIT V : Micro Electro Mechanical Systems (MEMS)

Syllabus: History, Effect of scaling, Fabrication Techniques: Oxidation, Physical Vapor Deposition, Chemical Vapor Deposition, Lithography, Etching, Wafer bonding, LIGA, DRIE, Applications: Lab on chip.

I. Introduction to MEMS

- **MEMS (Micro Electro Mechanical Systems)** are miniature devices (size 1 micron to 1 mm) that integrate **mechanical elements (structures), sensors, actuators and electronics on a common silicon substrate** using micro fabrication technology. Typical feature size: 1 - 100 microns; the whole device a few mm.
- Called **MST (Microsystems Technology)** in Europe and **Micromachines** in Japan.
- Mechanical parts (beams, diaphragms, gears, cantilevers) are made by **micromachining** (selective etching and depositing of layers), electronics by standard **IC (integrated circuit) processes** .
- **Components of a MEMS device:** microsensor, microactuator, microstructure (mechanical) and microelectronics (signal processing).

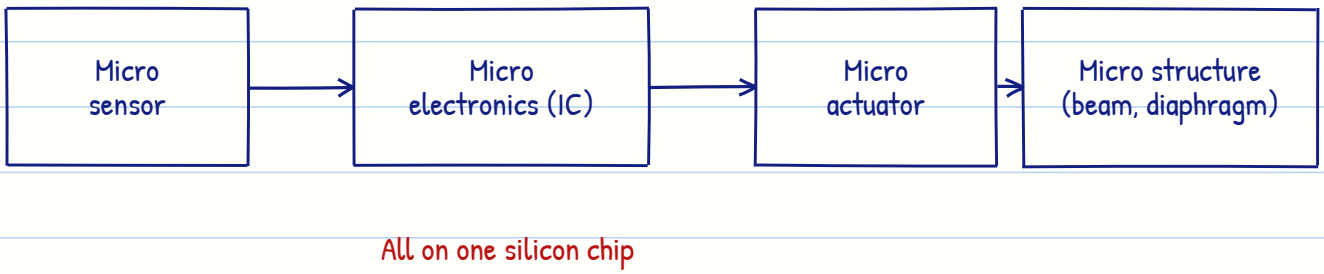


Fig: Elements of a MEMS device

Why silicon?

- High strength to weight, **Young's modulus 190 GPa (like steel)** , high hardness, **no mechanical hysteresis** , good thermal conductivity, **piezoresistive property** , mature IC technology, cheap single crystal wafers, can be bulk and surface micromachined. Other: SiO2, Si3N4, polysilicon, quartz, SiC, polymers (PDMS, SU-8), glass, metals (Au, Al, Ni).

Advantages

- Small size and weight, **low cost in batch fabrication** , low power, high speed and sensitivity, integration with electronics, high reliability, new functions.

Limitations

- High initial cost of the clean room / equipment, packaging is difficult and costly, limited design standard, small forces, **stiction** (sticking of surfaces), difficult testing and calibration.

2. History of MEMS

Year	Event
1947	Invention of transistor (Bell Labs) - base of silicon technology
1954	Piezoresistive effect in silicon and germanium discovered by C.S. Smith
1958	First integrated circuit (Jack Kilby)
1959	Richard Feynman lecture "There's plenty of room at the bottom" - idea of micro machines
1960s	Silicon strain gauges and pressure sensors
1967	Resonant gate transistor (Harvey Nathanson, Westinghouse) - first surface micromachined device
1970s	Bulk micromachined silicon pressure sensors (Kulite, National Semiconductor), anisotropic etching of silicon
1979	Inkjet printer nozzle (HP), micro gas chromatograph (Stanford)

1982	Kurt Petersen paper "Silicon as a mechanical material", LIGA process developed (Germany)
1987-88	Polysilicon micro motors, gears at UC Berkeley, term MEMS used
1990s	ADXL accelerometer (Analog Devices) for air bags, Digital Micromirror Device (Texas Instruments)
2000s	MEMS gyroscopes, microphones, RF MEMS, BioMEMS, lab on chip, MEMS in mobile phones
2010s on	Wearables, IoT, nanoelectromechanical systems (NEMS), MEMS microphones, sensor fusion in smart phones

3. Effect of Scaling

- **Scaling** studies how the physical quantities change when the size of a system is reduced by a scale factor **S** (a length L is reduced to S x L). The **volume** goes as S^3, **area** as S^2, and length as S.
- Different laws of physics dominate at the micro scale - **surface effects dominate volume effects.**

Geometric scaling

- Area ~ L^2 (S^2), Volume ~ L^3 (S^3). **Surface area / Volume ratio ~ 1 / L** - increases as size reduces. Eg: for a cube of L = 1 m ratio = 6, for L = 1 micron ratio is 6 x 10^6.
- So **mass, inertia and weight (~ L^3) fall faster than surface forces (~ L^2)** like friction, adhesion, surface tension, heat transfer, viscosity.

Scaling in different domains

Quantity	Scaling	Meaning at micro scale
Mass / weight / inertia	L^3	Becomes negligible, gravity is unimportant
Surface area	L^2	Surface forces dominate

Friction and adhesion (stiction)	L^2 (compared to L^3)	Parts stick, friction more important than inertia
Electrostatic force	L^2 (F ~ eA V^2 / 2d^2)	Strong, so electrostatic actuation is used
Magnetic force	L^4 (or L^3)	Weak - magnetic actuators are not suitable
Heat capacity	L^3	Very fast heating / cooling
Heat loss (conduction)	L (area/length)	Large heat loss compared to heat stored
Natural frequency	1/L	Very high frequency (MHz), fast response
Stiffness	L	Spring constant falls with L
Power density	1/L	Higher power per unit volume

- **Time scaling:** time for motion t ~ L, so micro devices respond very quickly. **Acceleration** a ~ 1/L (a = 2 s / t^2), **velocity** ~ L.
- **Fluid effects:** viscous forces dominate, flow is **laminar (very low Reynolds number)** - mixing only by diffusion. Surface tension and capillary forces are strong - used in lab on chip.
- **Electrostatic vs magnetic actuation:** electrostatic force scales as L^2, magnetic as L^3 to L^4, so for micro devices **electrostatic (comb drives), piezoelectric and thermal** actuators are used rather than electromagnetic.

Importance of scaling

- Helps to decide which physical principle is best for a micro device; shows **why a design cannot be just a miniature copy** of the macro design.

4. MEMS Fabrication Techniques

- Made with the same steps as the IC process on a **silicon wafer** (single crystal, 100 - 300 mm diameter, 0.5 mm thick) in a **clean room** (class 100 or better).

The main steps: **deposition of thin films, patterning by lithography, and etching** - repeated for each layer.

- Three families: **Bulk micromachining** (etching into the wafer), **Surface micromachining** (depositing and etching thin layers on the wafer surface - using a **sacrificial layer**), and **LIGA / high aspect ratio** processes.

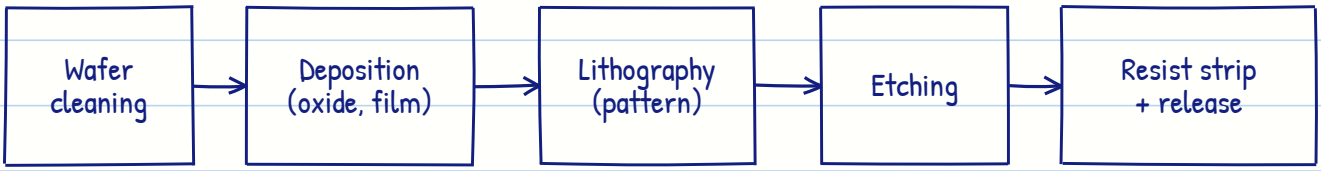


Fig: Basic MEMS process flow

4.1 Oxidation

- Growing a layer of **silicon dioxide (SiO2)** on the silicon wafer by heating the wafer in oxygen or steam at 900-1200 deg C in a furnace. Silicon reacts with the oxidant.
- **Dry oxidation:** Si + O2 -> SiO2. Slow, **dense, high quality, thin oxide** (gate oxide). **Wet oxidation:** Si + 2 H2O -> SiO2 + 2 H2. Faster, thicker (0.5 - 2 micron), lower quality - used for masking and isolation.
- Oxide thickness is given by the **Deal-Grove model:** $x^2 + A x = B (t + tau)$. Initially linear ($x \sim t$), later parabolic ($x \sim \sqrt{t}$). About **46% of the oxide thickness consumes silicon** .
- **Uses:** masking layer for etching and diffusion, electrical insulation, **sacrificial layer** in surface micromachining, passivation, dielectric of capacitors.

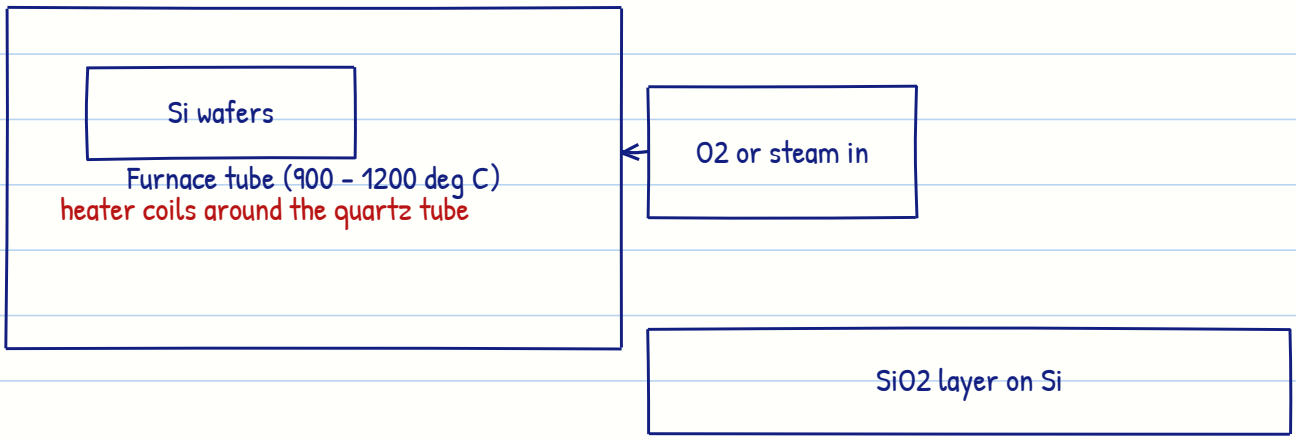


Fig: Thermal oxidation furnace

4.2 Physical Vapor Deposition (PVD)

- A **physical process** in which a solid source material is converted to vapour in vacuum and condenses as a thin film on the wafer. No chemical reaction. Film thickness from nanometres to microns. Used for **metals** (Al, Au, Cr, Ti, Pt).

(a) Evaporation

- The source is heated (**resistive or electron beam heating**) in a high vacuum (10^-6 torr) till it evaporates, and the vapour travels in straight line to the wafer (line of sight) - poor step coverage. Simple and fast.

(b) Sputtering

- A plasma of an inert gas (**Argon ions**) is created in the vacuum chamber. The ions are accelerated to the **target (source material, cathode)** ; by momentum transfer atoms are knocked out (sputtered) and deposit on the wafer (substrate). **DC sputtering** for metals, **RF sputtering** for insulators. **Magnetron sputtering** gives higher rate.
- **Advantages:** good adhesion, **good step coverage** , can deposit alloys and compounds with the same composition as the target, uniform films, low temperature.

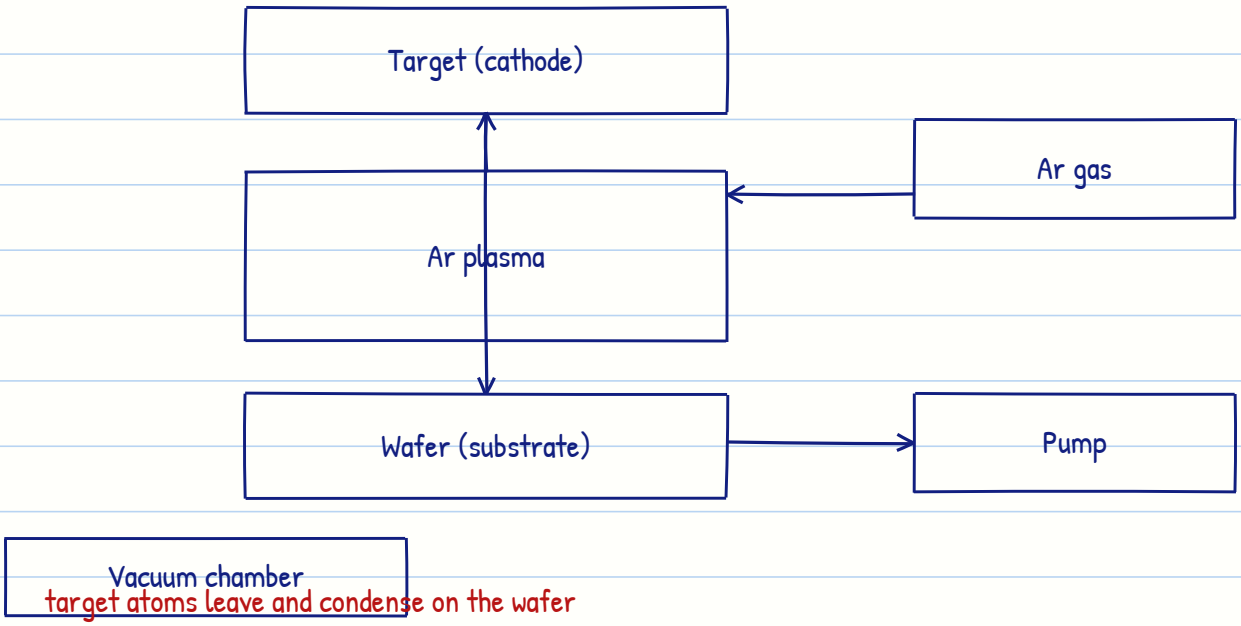


Fig: Sputtering process

4.3 Chemical Vapor Deposition (CVD)

- Thin film is deposited by a **chemical reaction of gases (precursors)** on the heated wafer surface. The by products are removed by the vacuum pump. Gives **dielectrics (SiO2, Si3N4), polysilicon, tungsten** and other films of good uniformity.
- **Steps:** reactant gas transported to the surface -> adsorption -> chemical reaction on the hot surface (film forms) -> by-product desorbs and is pumped out.
- Examples: **Polysilicon:** $SiH_4 \rightarrow Si + 2\ H_2$ (about 600 deg C). **Silicon nitride:** $3\ SiH_2Cl_2 + 4\ NH_3 \rightarrow Si_3N_4 + 6\ HCl + 6\ H_2$ (LPCVD, 800 deg C). **Silicon dioxide:** $SiH_4 + O_2 \rightarrow SiO_2 + 2\ H_2$ (at 400 deg C).

Types

- **APCVD** (atmospheric pressure) - simple, high rate, poor uniformity. **LPCVD** (low pressure, 0.1 - 1 torr) - **best uniformity and step coverage** , high temperature. **PECVD** (plasma enhanced) - **low temperature (200 - 350 deg C)** , can deposit on metals. Also **MOCVD** and **ALD** (atomic layer deposition).

Point	PVD	CVD
Process	Physical (evaporation / sputter)	Chemical reaction of gases
Material	Mostly metals	Dielectrics, poly-Si, metals
Temperature	Low	High (LPCVD), low (PECVD)
Step coverage	Poor (evap.) / good (sputter)	Very good, conformal
Vacuum	High vacuum	Low / atmospheric pressure
Safety	Safer	Toxic, flammable gases

4.4 Lithography

- **Lithography** transfers a **pattern** from a mask to the wafer. It is the most important step - decides the minimum feature size. **Photolithography** uses light and a light sensitive polymer called **photoresist** .

Steps of photolithography

- **1. Cleaning and dehydration bake** of the wafer. **2. Adhesion promoter (HMDS).**
- **3. Spin coating:** a few ml of photoresist is dropped on the wafer and spun at 1000 - 6000 rpm to get a uniform film (0.5 - 2 micron).
- **4. Soft bake** (90-100 deg C) to remove solvent.
- **5. Alignment and exposure:** the **mask (photomask - glass with chrome pattern)** is aligned and the wafer is exposed to UV light (365 nm i-line, 248 nm KrF, 193 nm ArF, EUV 13.5 nm). Types: contact, proximity and projection printing.
- **6. Development:** the exposed (positive resist) or the unexposed (negative resist) areas are dissolved in a developer solution.
- **7. Hard bake** to harden the resist -> then etching or deposition -> **8. Resist stripping** (acetone, O2 plasma).

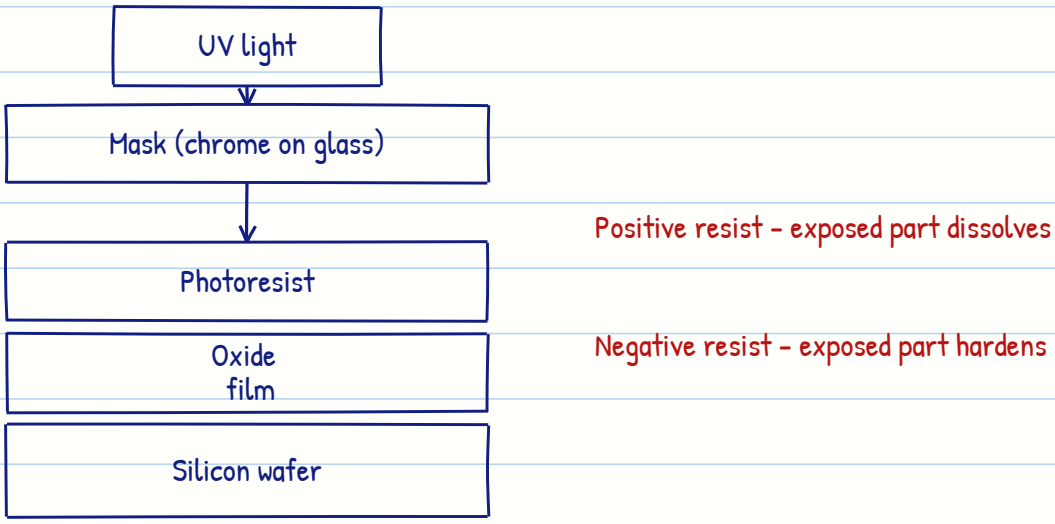


Fig: Photolithography

Positive vs negative resist

- **Positive:** the exposed region becomes soluble - the pattern is same as the mask; better resolution. **Negative:** exposed region becomes insoluble - pattern is the inverse of the mask; good adhesion, lower resolution; eg **SU-8** (thick negative resist for MEMS, LIGA like structures).
- **Other lithography:** electron beam (direct write, 10 nm, slow), X-ray, nanoimprint, **soft lithography (PDMS stamp).**

4.5 Etching

- **Etching** is the selective removal of material not protected by the mask, to create the structures (holes, grooves, membranes). Measures: **etch rate, selectivity** (ratio of etch rate of the material to the mask), **anisotropy** (directionality) and **aspect ratio** .

Wet etching

- Wafer dipped in liquid chemical etchants.
- **Isotropic etching:** equal rate in all directions - undercut below mask; **HNA (HF**

+ HNO3 + acetic acid) for silicon; **HF / buffered HF** for SiO2.

- **Anisotropic etching:** depends on the **crystal plane - KOH, TMAH, EDP** etch Si (100) plane fast, but the (111) plane very slowly (100x less). It forms **V-grooves and pyramidal pits with 54.74 deg walls** ; with (110) wafers gives vertical walls. **Etch stop** by heavy boron doping (p++) or an electrochemical etch stop.
- **Advantages:** simple, low cost, high selectivity, high rate. **Limitations:** limited shapes (crystal planes), undercutting, surface tension problems, not suitable for very small features.

Dry etching

- Uses **gases / plasma** in a vacuum chamber.
- **Plasma etching** (chemical, isotropic), **Reactive Ion Etching (RIE)** - plasma gives reactive ions (CF4, SF6) accelerated vertically toward the wafer: **physical bombardment + chemical reaction -> anisotropic** . **Sputter etching / ion milling** (purely physical, low selectivity). **Vapour phase etching** (XeF2 for Si, HF vapour for SiO2) for releasing structures without stiction.
- **Advantages:** anisotropic, vertical walls, any crystal orientation, high resolution. **Limitations:** expensive equipment, lower selectivity, slower.

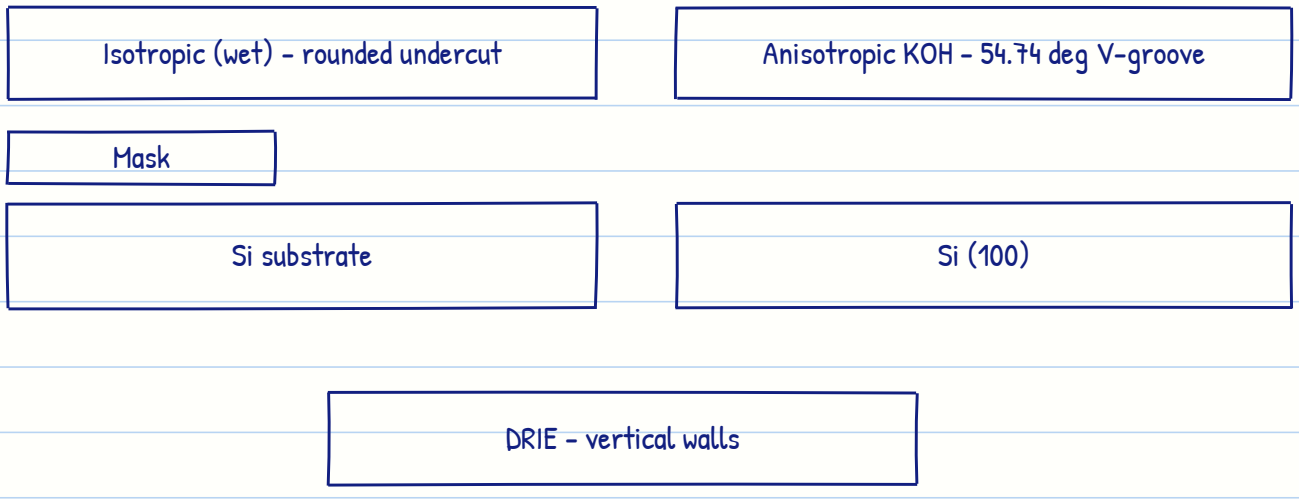


Fig: Types of etch profiles

Bulk and surface micromachining

- **Bulk micromachining:** removes silicon from the wafer by wet / dry etching to form diaphragms, cantilevers and cavities (pressure sensor, accelerometer). Wafer thickness is used.
- **Surface micromachining:** thin films are built on the surface: **sacrificial layer (SiO2 / PSG)** is deposited, then **structural layer (polysilicon)** is deposited and patterned, then the sacrificial layer is **etched away (release)** with HF to free the moving structure (comb drives, gears, cantilever). Care for stiction - use **critical point drying** .

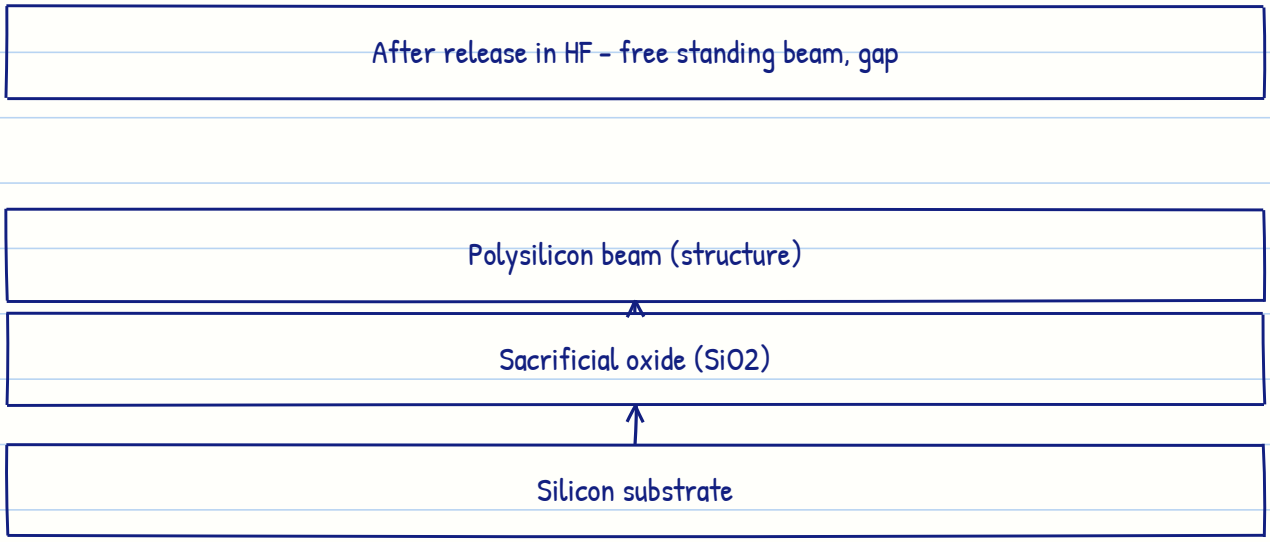


Fig: Surface micromachining - deposit, pattern, and release of a polysilicon cantilever

4.6 DRIE (Deep Reactive Ion Etching)

- A highly anisotropic dry etching to get **deep (100s of microns), vertical walls with a high aspect ratio (> 20:1, up to 100:1)** in silicon. **Bosch process (Robert Bosch, 1990s)** - alternate cycles of:
  - **Etch step:** SF6 plasma isotropically etches silicon a little (few microns).
  - **Passivation step:** C4F8 plasma deposits a **fluorocarbon polymer (Teflon like)** layer on all surfaces.
  - Next etch step: ions bombard the bottom and **remove polymer from the**

floor only (not from the side walls), so the side walls are protected and the etching continues downward. Cycle repeats (**scalloped walls**).

- **Cryogenic process:** wafer cooled to -110 deg C with SF6 + O2 for smooth walls.
- **Selectivity** to oxide mask 200:1 and to photoresist 75:1. Etch rate 1 - 20 micron / min.
- **Uses:** through wafer vias, deep trenches, MEMS accelerometers, gyroscopes, microfluidic channels, comb drives, SOI wafers (oxide as etch stop).

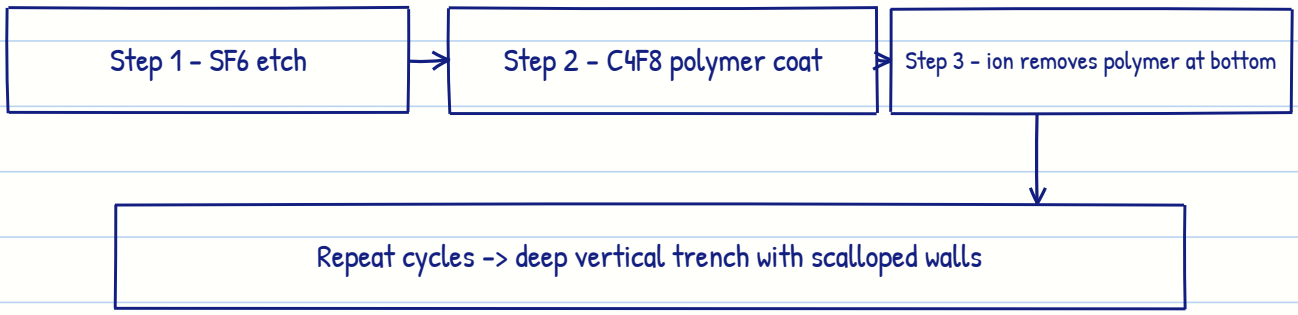


Fig: Bosch process cycle (DRIE)

4.7 Wafer Bonding

- Joining two wafers permanently to make **sealed cavities, multilayer structures, and for packaging** (capping of MEMS), where the single wafer micromachining is not enough.
- **Direct (fusion) bonding:** two very clean, flat, hydrophilic Si / SiO2 wafers are pressed (room temp.), then **annealed at 800 - 1100 deg C** - strong covalent Si-O-Si bonds. Used for **SOI wafers** .
- **Anodic (electrostatic) bonding:** silicon + **Pyrex (borosilicate) glass** heated to **300 - 450 deg C** and a **voltage of 500 - 1000 V** is applied (Si +ve, glass -ve). Na+ ions in glass drift away, creating an electrostatic field and an oxide bond at the interface. Hermetic seal - used for pressure sensors and microfluidics.

- **Intermediate layer bonding: eutectic** (Au-Si, Au-Sn), **glass frit** , **thermo compression** , adhesive (polymer, SU-8, BCB) - lower temperature, bond even with metal / CMOS present.
- **Requirements:** flat, clean, smooth surfaces (roughness < 1 nm), particle free, void free bond; **thermal expansion match** to avoid stress.

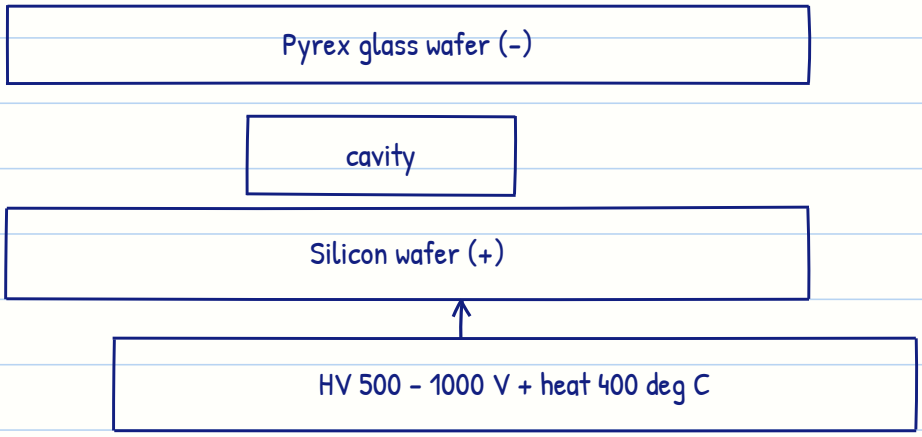


Fig: Anodic bonding of silicon to glass

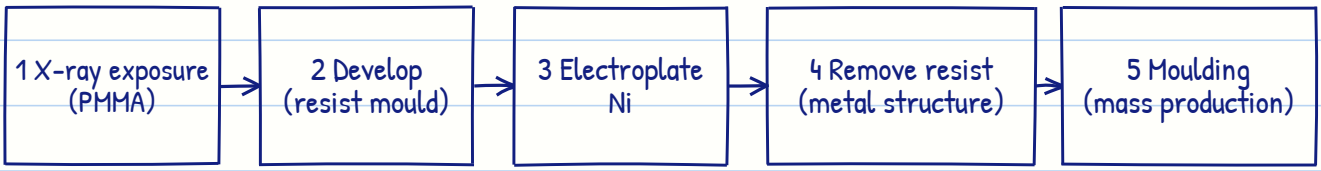
4.8 LIGA Process

- **LIGA** - German acronym: **LI** thographie (lithography), **G** alvanoformung (electroplating) and **A** bformung (moulding). Developed at Karlsruhe, Germany in the early 1980s (Becker and Ehrfeld). Makes **high aspect ratio (up to 100+), 3D metal / plastic / ceramic microstructures** with heights from tens microns to several mm, not limited to silicon.

Steps

- **1. X-ray lithography:** a thick layer (up to 1 mm) of **PMMA (polymethyl methacrylate)** resist on a conductive substrate is exposed through an X-ray mask using **synchrotron X-ray** radiation. The exposed PMMA gets weaker (chain scission).
- **2. Development:** the exposed PMMA is removed - a deep resist mould with very vertical walls.

- **3. Electroplating (Galvanoformung):** metal (**nickel, copper, gold**) is electro deposited into the mould cavities.
- **4. Resist removal:** the PMMA is dissolved - a free standing **metal microstructure** , or used as a **mould insert (mould tool)** .
- **5. Moulding (Abformung):** the metal mould insert is used for **mass production by injection moulding / hot embossing / casting** of plastic parts.



Synchrotron X-ray -> deep and vertical walls

Fig: LIGA process

Variants

- **UV-LIGA / SU-8:** uses UV light with SU-8 resist (cheaper, no synchrotron, height to 1 mm) and **sacrificial LIGA** .
- **Advantages:** very high aspect ratio, vertical smooth walls (roughness < 50 nm), wide range of materials, mass production by moulding. **Limitations:** very **costly** (synchrotron source, X-ray mask), long process, few facilities, mostly not compatible with IC electronics.
- **Applications:** micro gears, micro motors, microfluidic channels, accelerometers, optical switches, fiber optic connectors, micro nozzles, spinnerets.

4.9 Comparison of the micromachining methods

Point	Bulk	Surface	LIGA

Material	Silicon wafer	Thin films on wafer	Metal, plastic, ceramic
Structure size	Hundreds of microns thick	Few microns thick	Up to mm tall
Aspect ratio	Medium (DRIE high)	Low	Very high
Cost	Medium	Medium	High
IC compatibility	Good	Very good	Poor
Eg	Pressure sensor, nozzle	Comb drive, gyroscope	Micro gears, moulds

5. Applications of MEMS

- **Automotive:** air bag accelerometer (ADXL), tyre pressure (TPMS), manifold absolute pressure (MAP), gyroscope for ESC, fuel injection nozzle.
- **Consumer / mobile:** accelerometer and gyroscope (screen rotation, games), MEMS microphone, compass (magnetometer), inkjet print head, **DMD (digital micromirror device) projectors** .
- **Medical (BioMEMS):** disposable blood pressure sensor, pacemaker sensor, micro needles for drug delivery, lab on chip, DNA chips, hearing aid, endoscope.
- **Aerospace / defence:** inertial navigation (gyros), micro satellites, RF MEMS switches, drones.
- **Industrial:** vibration, pressure, flow and gas sensors, optical MEMS switches (fibre optic communication), micro mirrors, **RF filters** , energy harvesters.

6. Lab on Chip (LOC)

- A **lab on a chip** is a device of a few mm to cm size that integrates **one or many laboratory functions (sample preparation, mixing, reaction, separation, detection) on a single chip** using **microfluidics** - handling tiny volumes of fluid (nano to picolitre) in channels of 10 - 100 micron width. Also called **micro Total Analysis System (uTAS)** .
- **Principle:** at micro scale the flow is **laminar** (low Reynolds number), surface

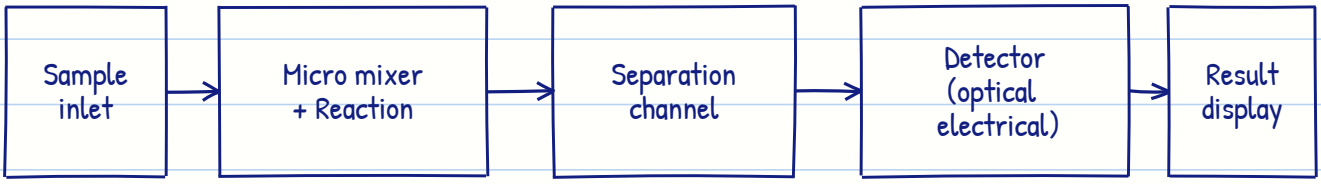
forces dominate, diffusion is fast (short distances) - so mixing is by diffusion; fluid movement is by **pressure, capillary action, electro-osmosis, electrophoresis, electrowetting** or micro pumps.

Main parts

- **Sample inlet / reservoirs, microchannels, micro pumps and valves, micro mixers, reaction chambers, separation unit** (capillary electrophoresis), **detectors** (optical fluorescence, electrochemical), **heaters, electrodes** , and a readout electronics.

Materials and fabrication

- **Glass and silicon** (etching, anodic bonding), and **polymers - PDMS (soft lithography), PMMA, SU-8, PC, COC** (hot embossing, injection moulding, laser cutting). **Paper based** low cost chips. Channels are sealed by bonding a cover plate.



Pumps, valves and electrodes on the same chip

Fig: Lab on chip - functional blocks

Advantages

- **Very small sample and reagent volume,** fast analysis (minutes), low cost per test, **portable and point of care** use, automation and parallel testing, low power and waste, high sensitivity and safe.

Limitations

- Detection at small volumes is difficult, clogging of channels and bubbles, the

cost of the fabrication / packaging, interface with the macro world (tubing), standardisation and reliability issues.

Applications

- **Medical diagnostics** - glucose monitoring, pregnancy test, blood analysis, **COVID / infectious disease tests,** DNA / PCR chips, drug discovery and delivery, **organ on chip** ; environmental monitoring (water testing), food safety, forensic, chemical synthesis (micro reactor), and biological cell sorting and counting.

Important Questions (Unit V)

- What is MEMS? Explain the history of MEMS and the advantages of MEMS.
- Explain the effect of scaling on MEMS devices (geometric, electrostatic, heat, fluid).
- Explain oxidation and the PVD / CVD processes used in MEMS fabrication.
- Explain the photolithography process with a neat sketch.
- Explain wet and dry etching; bulk and surface micromachining; DRIE (Bosch process).
- Explain wafer bonding techniques and the LIGA process.
- Write a short note on Lab on chip and applications of MEMS.